Gathmann CA Solutions

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1 Exercise 0.15(a,d)

Let $f \in \mathbb{R}[x]$ be a polynomial vanishing on $\mathbb{Z}$. Then, f is the zero polynomial, so it vanishes everywhere. Then, $\mathbb{Z}$ can't be a variety since then it would be bijective to $\mathbb{R}$, but the cardinalities don't match. The same argument works for part (d) as well.

2 Exercise 0.15(b)

The coordinate ring of $\mathbb{R}\backslash\{0\}$ is $\mathbb{R}[x]$, so it's not a variety.

3 Exercise 0.15(c)

4 Exercise 0.15(e)

5 Exercise 1.8

Let $x \in I_1 \cap \cdots \cap I_n$. We'd like to show that $x \in I_1 \cdots I_n$.

Let's first show it for $n = 2$. Since I, J are coprime, there exists $a_1, a_2 \in R$ and $z_1 \in I$ and $z_2 \in J$ such that $a_1 z_1 + a_2 z_2 = 1$ by the partition of unity.

Then, $x = x(a_1 z_1 + a_2 z_2) = x a_1 z_1 + x a_2 z_2$.

Notice that both terms are in the product, so we're done.

Now, assume the statement holds for $n - 1$. Then, we have that z is in $I_1 \cdots I_{n-1}$.

Let's now show that $I_1 \cdots I_{n-1}$ and I_n are coprime. This is sufficient since we can apply the $n = 2$ case to finish the proof.

6 Exercise 1.9

Let $I = \langle x_1, \ldots, x_n \rangle$.

Let $J = \langle x_1^2, \ldots, x_n^2 \rangle$.

Consider $R = K[x_1, \ldots, x_n]/J$. Since I contains J, the image of I is nonzero in R. Then, a generator for I in $K[x_1, \ldots, x_n]$ would be a generator for I in R. But I is now a subspace of the n-dimensional vector space R over K, so we need n generators.

7 Exercise 1.11(a)

Let J be an ideal in $\mathbb{C}[x]$. Since $\mathbb{C}[x]$ is a PID, $J = (f)$. Let $f = \pi_{i=1}^{n}(x - a_i)$. If J is a radical ideal, f is separable over $\mathbb{C}$. Thus, the vanishing ideal is simply the roots of f.

Recall that $\mathbb{A}^1_{\mathbb{C}}$ has the cofinite topology as its Zariski topology. Thus, to every subvariety, we can assign the ideal generate by the polynomial that has the subvariety as its roots.

These two assignments are clearly inverses of each other.

8 Exercise 1.11(b)

The vanishing locus of (x^2+1) is empty, but it's not a unit ideal.

9 Exercise 2.11(a)

Let $f \in R/I_a$ be non-zero.

Then, $f(a) \neq 0$ and f is continuous.

Then, f is non-zero around a small enough neighborhood of a.

10 Exercise 2.11(b)

Let $f_1, \ldots, f_n \in R$ such that $V(f_1, \ldots, f_n) = \varnothing$.

Then, there's no point of $[0,1]$ where all the functions vanish. Then, $f_1^2 + \cdots + f_n^2$ is a positive function that doesn't vanish at any point of $[0,1]$.

11 Exercise 2.24

Let P, Q be distinct maximal ideals of a ring R.

Assume P contains Q^m for some $m \geq 1$.

Since P is maximal, it's prime, and it's thus radical.

Thus, taking the radicals of $Q^m \subset P$, we get $Q \subseteq P$, which is a contradiction.

12 Exercise 7.22(a)

Recall that irreducible elements correspond to maximal principal ideals.

Let $r \in R$. If r is irreducible, we're done.

Otherwise, $r = r_1 r_2$ for some $r_1, r_2 \in R$. If r_1 and r_2 are irreducible, we're done. Otherwise, assume without loss of generality that r_1 is reducible. Notice that $r_1 \mid r$ if and only if $(r) \subseteq (r_1)$. Thus, this process has to stop at some point since otherwise we'll produce an infinite ascending chain of ideals.

13 Exercise 7.22(b)

Every ideal I is finitely generated, so assume $\sqrt{I} = \langle x_1, \ldots, x_n \rangle$.

For every x_i, $x_i^{k_i} \in I$ for some $k_i \geq 1$.

Let $k = \sum_{i=1}^{n} \{k_i\}$.

Consider $\sqrt{I}^k$. Then, using the binomial formula, we have the desired result.

14 Exercise 7.23

This is an immediate consequence of Proposition 6.7(b). In particular, considering the contraction of any ascending (descending) chain of ideals, and then extending back, we get the desired result.